Rosen — §8.6: Applications of Inclusion–Exclusion

Selected Exercises

Alston

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Exercise 5

Find the number of primes less than 200 using the principle of inclusion–exclusion.

Exercise 6

An integer is called *squarefree* if it is not divisible by the square of a positive integer greater than 1. Find the number of squarefree positive integers less than 100.

Exercise 8

How many onto functions are there from a set with seven elements to one with five elements?

Exercise 14

What is the probability that none of 10 people receives the correct hat if a hatcheck person hands their hats back randomly?

Exercise 17

How many ways can the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 be arranged so that no even digit is in its original position?

Exercise 18

Use a combinatorial argument to show that the sequence $\{D_n\}$, where D_n denotes the number of derangements of n objects, satisfies the recurrence relation

$$D_n = (n-1)(D_{n-1} + D_{n-2})$$

for $n \geq 2$. [*Hint*: Note that there are $n-1$ choices for the first element k of a derangement. Consider separately the derangements that start with k that do and do not have 1 in the kth position.]

Exercise 19

Use Exercise 18 to show that

$$D_n = nD_{n-1} + (-1)^n$$

for $n \geq 1$.

Exercise 23

Use the principle of inclusion–exclusion to derive a formula for $\varphi(n)$ when the prime factorization of n is $n = p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m}$.

Exercise 24

Show that if n is a positive integer, then

$$n! = \binom{n}{0} D_n + \binom{n}{1} D_{n-1} + \cdots + \binom{n}{n-1} D_1 + \binom{n}{n} D_0,$$

where D_k is the number of derangements of k objects.

Exercise 27

Recall (Theorem 1): the number of onto functions from a set with m elements to a set with n elements is

$$n^m - \binom{n}{1}(n-1)^m + \binom{n}{2}(n-2)^m - \cdots + (-1)^{n-1}\binom{n}{n-1} \cdot 1^m.$$

Prove Theorem 1.